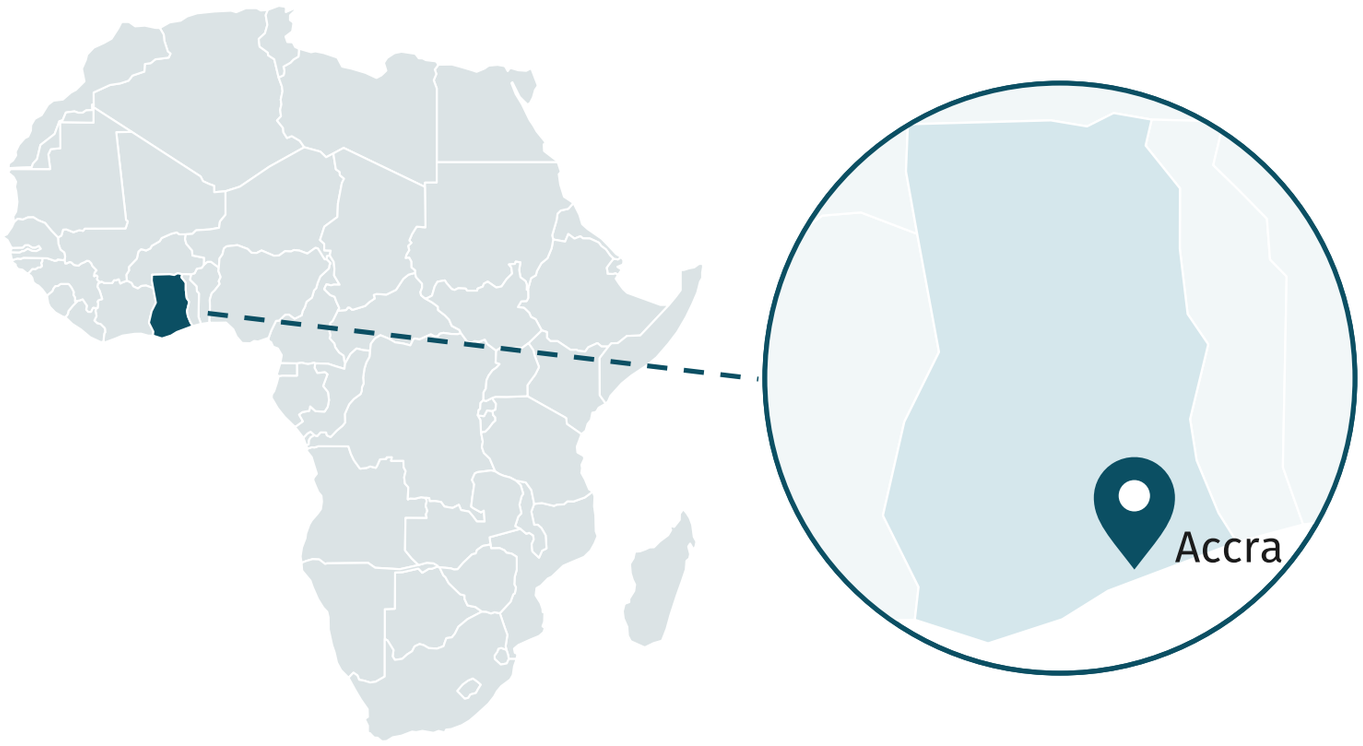
Drug-resistance and vaccine-target surveillance using nanopore sequencing, Ghana

A genomic surveillance study was conducted to monitor antimalarial drug resistance markers and vaccine-target diversity in Ghana.



Samples analysed

196

109 prospective VWB (2022)
87 retrospective DBS (2018)

Sites

Ledzokuku-Krowor Municipal Assembly (LEKMA) Hospital, Accra and three facilities in and around **Navrongo**

Priority use case

Antimalarial drug-resistance surveillance,
with secondary surveillance of vaccine-target diversity.

Approach

End-to-end targeted nanopore workflow on VWB and DBS — multiplexed amplicon sequencing of *pfcrt*, *pfdhfr*, *pfdhps*, *pfmdr1*, *pfkelch13*, and *pfcsp*, analysed locally with the Nextflow-based nano-rave pipeline.

Samples

109 prospective leucocyte-depleted VWB collected August–September 2022 and 87 retrospective DBS from 2018, across **Ledzokuku-Krowor** Municipal Assembly (LEKMA) Hospital in Accra and three facilities in and around **Navrongo**.

Key findings

Over 99% of samples inferred chloroquine-susceptible (1 of 196 carried *pfcrt* K76T); *pfdhfr* S108N in 93.4% and *pfdhps* A437G in 90.3%; nine *pfkelch13* mutations found, none previously associated with artemisinin partial resistance; multiple *pfcsp* C-terminal SNPs above 50% frequency relative to the vaccine reference.

Action supported

Standardised, end-to-end genomic surveillance can be implemented locally using DBS-compatible workflows; persistent sulfadoxine–pyrimethamine resistance markers were detected, but no validated artemisinin-resistance mutations.

Relevance to this section

Targeted long-read nanopore sequencing implemented locally, with portable instrumentation and laptop-compatible analysis.

Full record — ethics, sampling, library preparation, bioinformatics, complete marker frequencies and metadata fields: Case Study 1, Section 13.